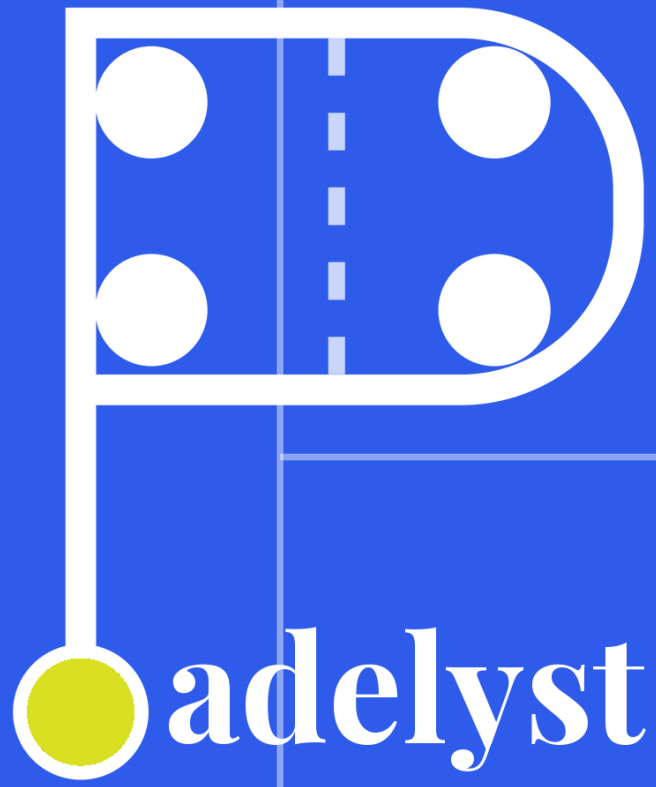



# Shot recognition in padel match videos

**Applied Data Science Project, Checkpoint 1**

**Alessandro de Stasi, Gianbattista Busonera, Gabriele Tebano**

Mentor: Giacomo Blanco, LINKS Foundation

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CONTEXT

# Padel is one of the fastest-growing sports in the world

**91,234**

padel courts worldwide

August 2026

**~14,000**

new courts in one year

worldwide

**~10,000**

courts in Italy

second only to Spain

**417**

courts in the province of Turin

138 clubs

Sources: FIP, World Padel Report 2026 (courts worldwide and by country). FIP Research Department, via Quotidiano Piemontese, 29 Sep 2025 (province of Turin).

THE PROBLEM

# The feedback is in the video, but no amateur rewinds a whole match

**937**

**shots in 22 minutes of play, across two finals**

All the labelled footage there is. Our count, PadelTracker100

1. **Slow**
   Finding one smash means rewinding an hour of footage
2. **Tedious**
   Turning it into statistics by hand is even worse
3. **Lost**
   So amateurs skip it, and lose the feedback entirely

VALUE PROPOSITION

For padel players and coaches tired of rewinding, Padelyst is your padel analyst that finds and names every shot of a match, so training starts from what really happened on court.

## FROM VIDEO TO SHOTS

Available today

1. Match video
2. Ball

This project

3. Players
4. Poses
5. When: a shot
6. Which: its type

# We start from a match video

One fixed camera, the broadcast view of a professional final

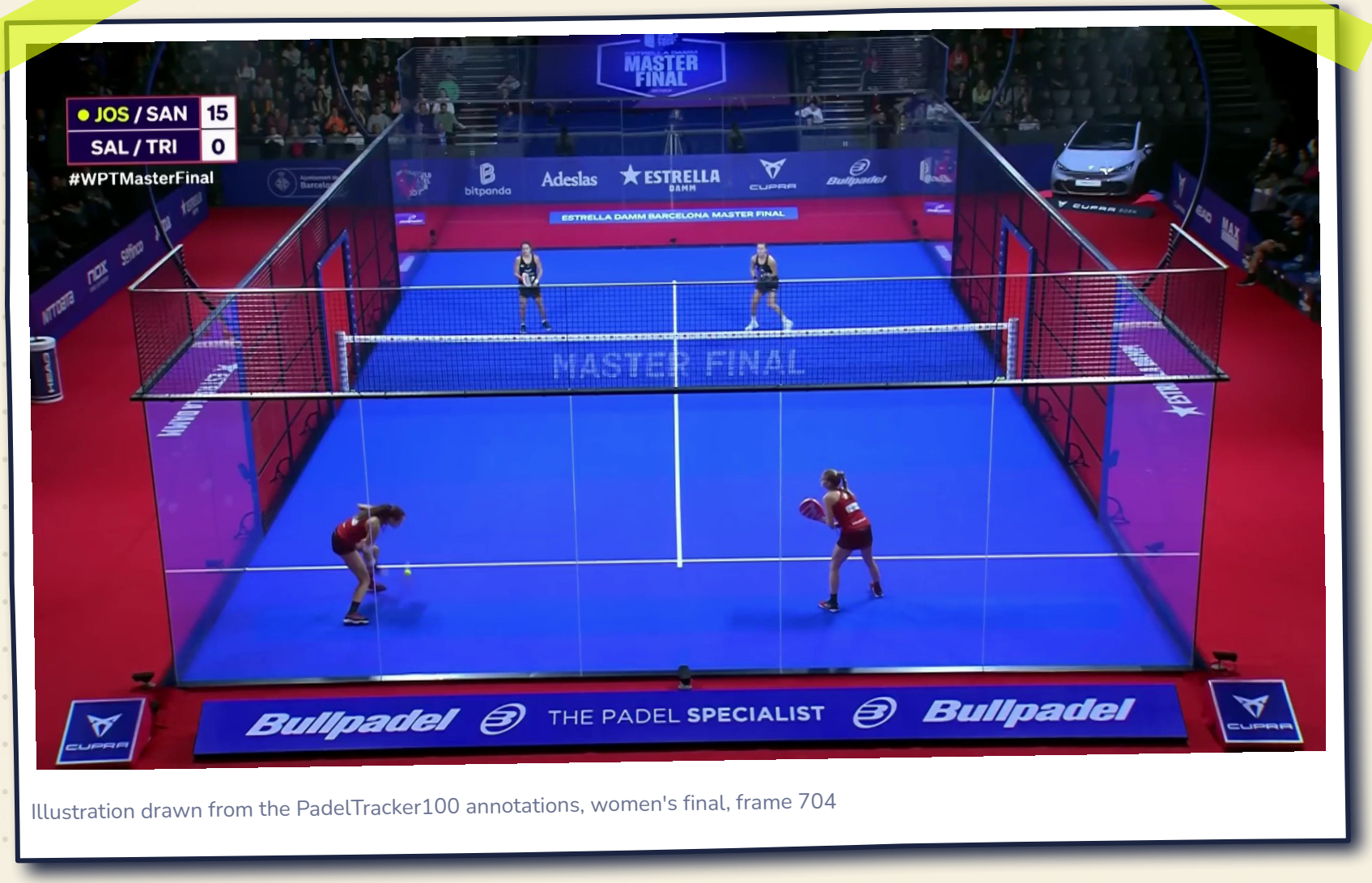



Illustration drawn from the PadelTracker100 annotations, women's final, frame 704

## FROM VIDEO TO SHOTS

Available today

1. Match video
2. **Ball**

This project

3. Players
4. Poses
5. When: a shot
6. Which: its type

# The ball is tracked

Given: where the ball is, frame by frame (last year's project)




Illustration drawn from the PadelTracker100 annotations, women's final, frame 704

**FROM VIDEO TO SHOTS**

Available today

1. Match video
2. Ball

This project

3. **Players**
4. Poses
5. When: a shot
6. Which: its type

# We find the players

A bounding box for each of the four players

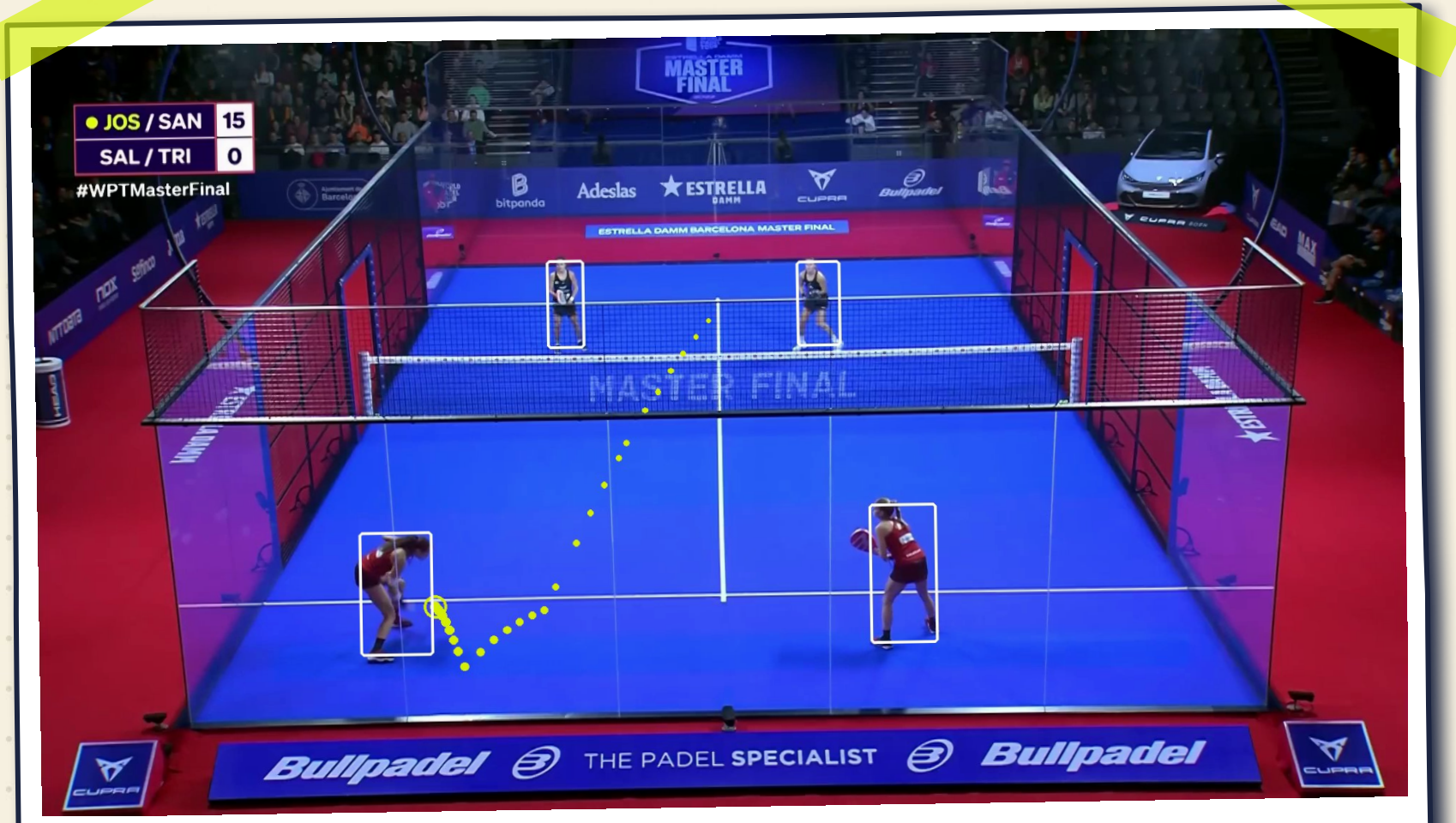



Illustration drawn from the PadelTracker100 annotations, women's final, frame 704

## FROM VIDEO TO SHOTS

Available today

1. Match video
2. Ball

This project

3. Players
4. **Poses**
5. When: a shot
6. Which: its type

# We estimate their poses

Body keypoints per player: arms, legs, torso

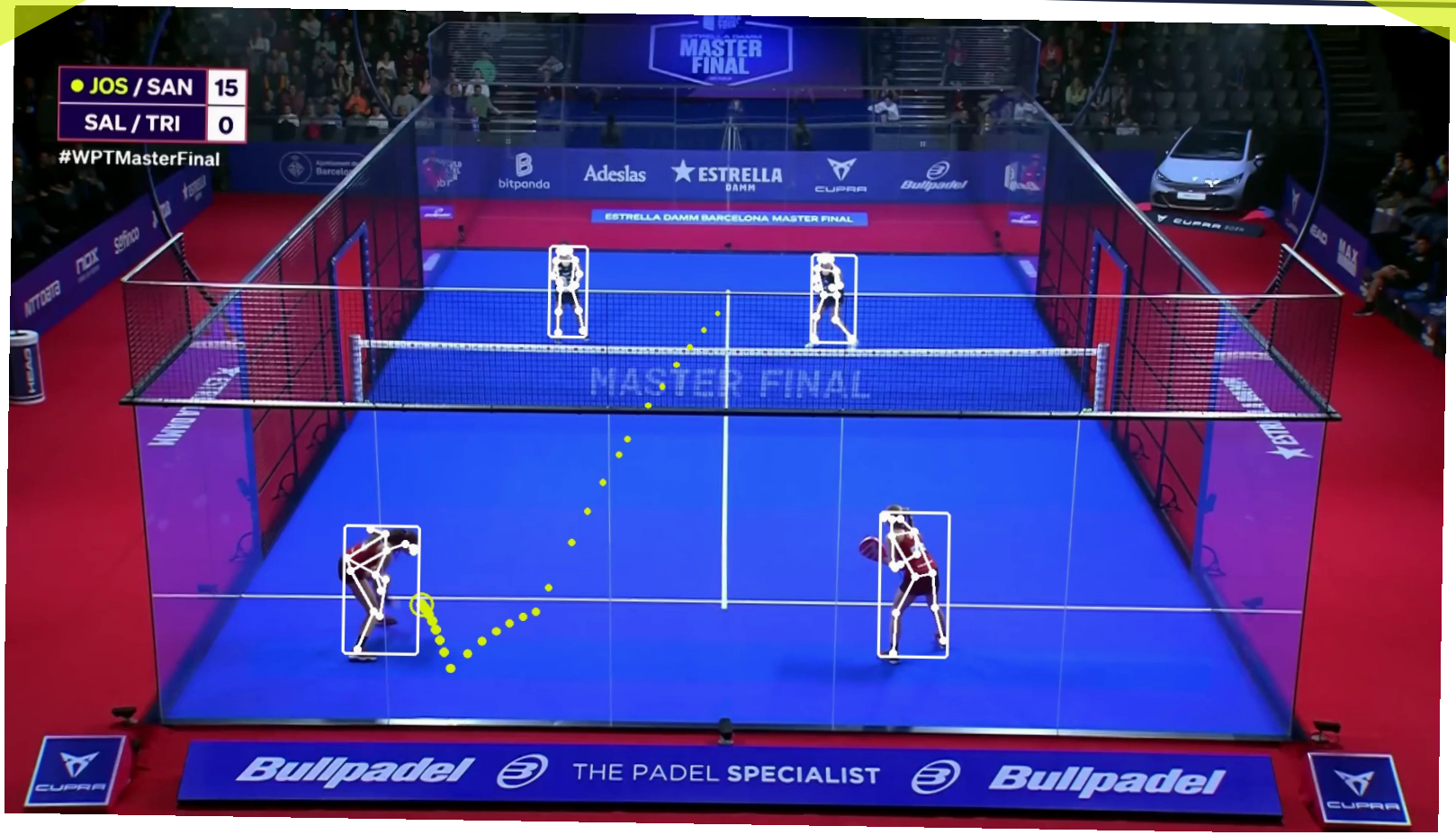



Illustration drawn from the PadelTracker100 annotations, women's final, frame 704

**FROM VIDEO TO SHOTS**

Available today

1. Match video
2. Ball

This project

3. Players
4. Poses
5. **When: a shot**
6. Which: its type

# When is a shot played?

Find the moment of every shot in the match

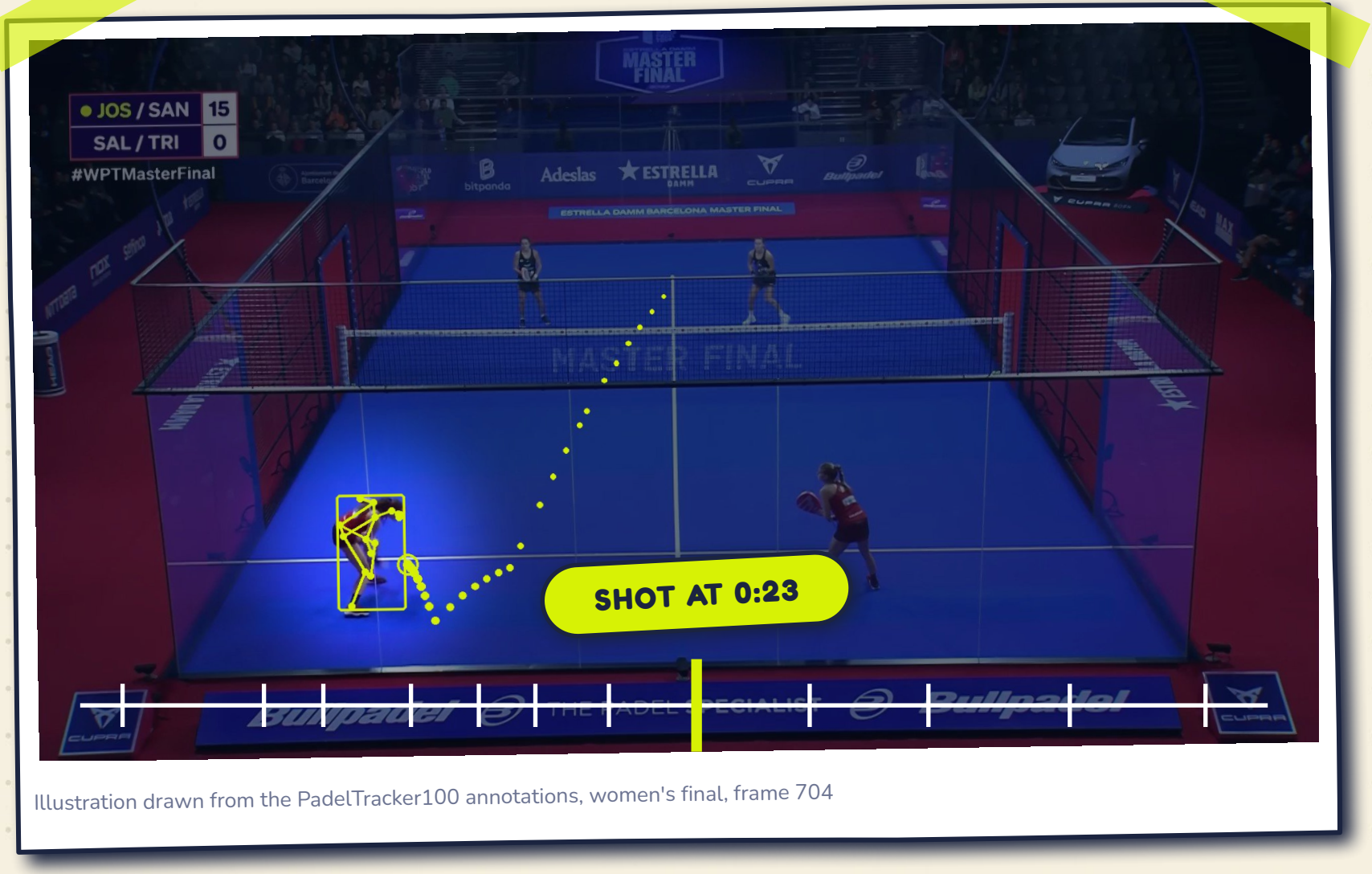



Illustration drawn from the PadelTracker100 annotations, women's final, frame 704

## FROM VIDEO TO SHOTS

Available today

1. Match video
2. Ball

This project

3. Players
4. Poses
5. When: a shot
6. **Which: its type**

# Which shot is it?

One of six: forehand, backhand, smash, serve, dropshot, other

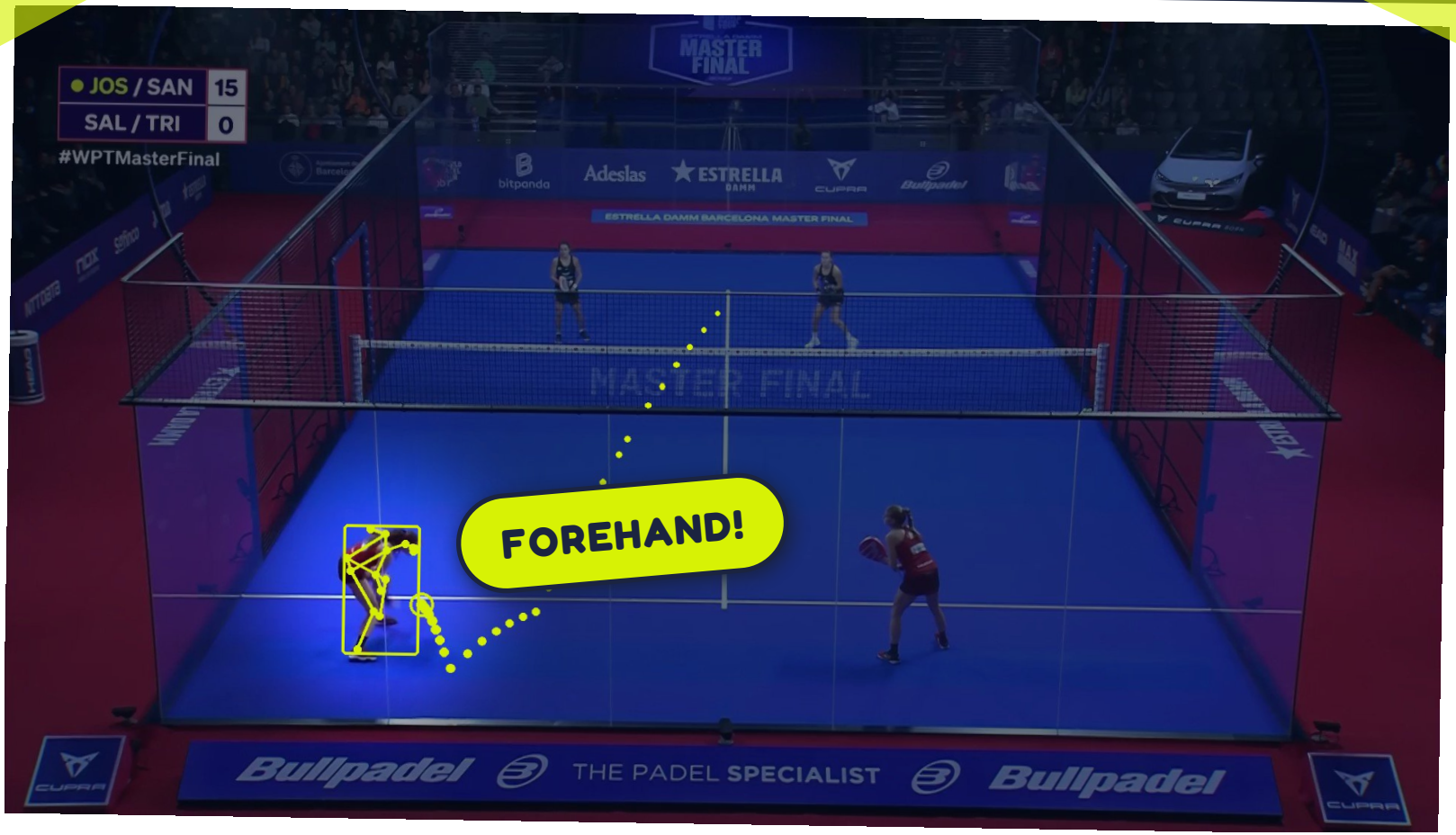
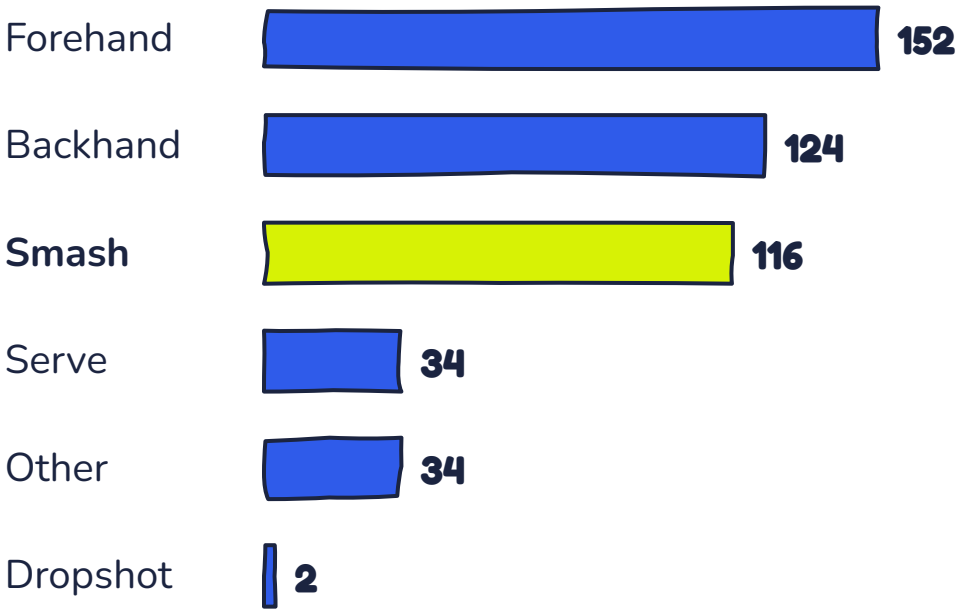



Illustration drawn from the PadelTracker100 annotations, women's final, frame 704

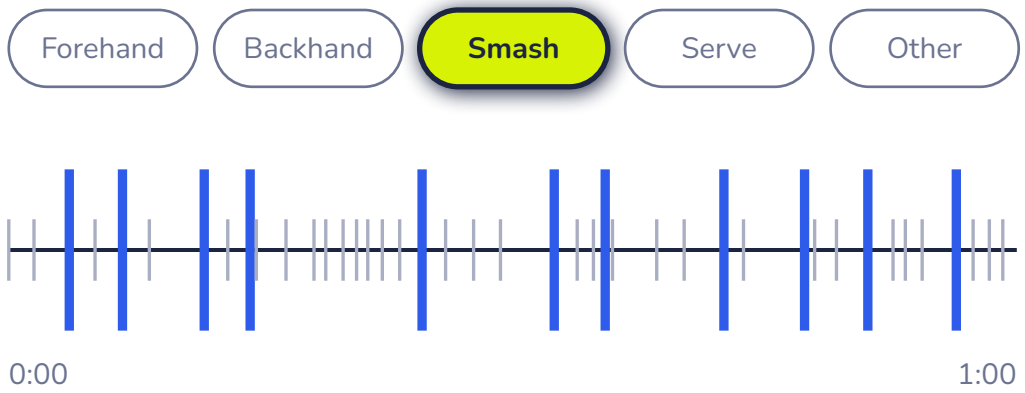
THE SOLUTION

# Found and named, the shots turn into statistics

### Shots per type, women's final




### The first minute, filtered on smash



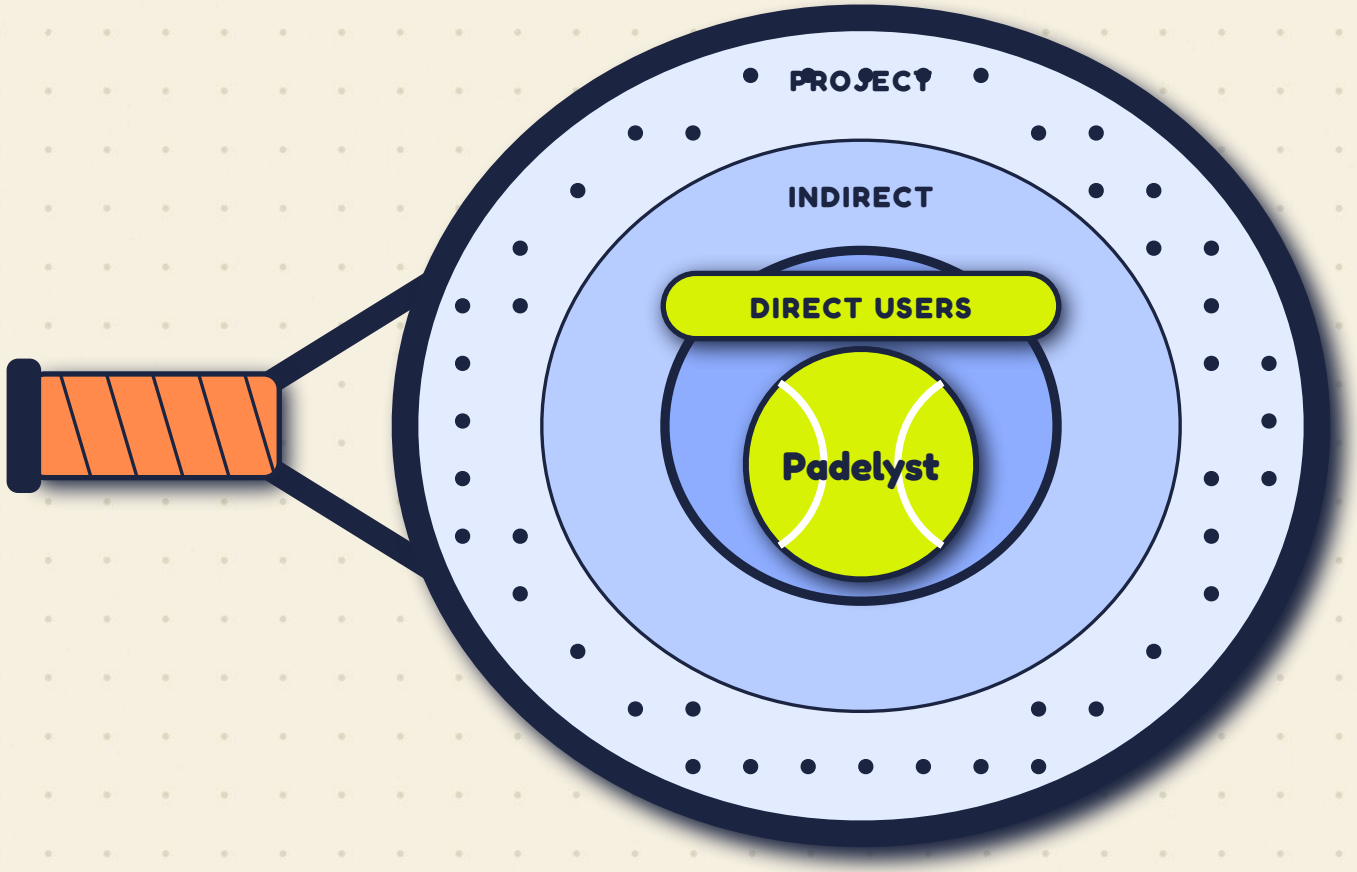

**11 smashes in one minute**

Each one a click away: no rewinding.

Real labels from the women's final (PadelTracker100), used here as a stand-in for the output of our model. Our count.

STAKEHOLDERS

# Players, coaches and clubs are our direct users




## Direct users

### Players
See their own game, shot by shot

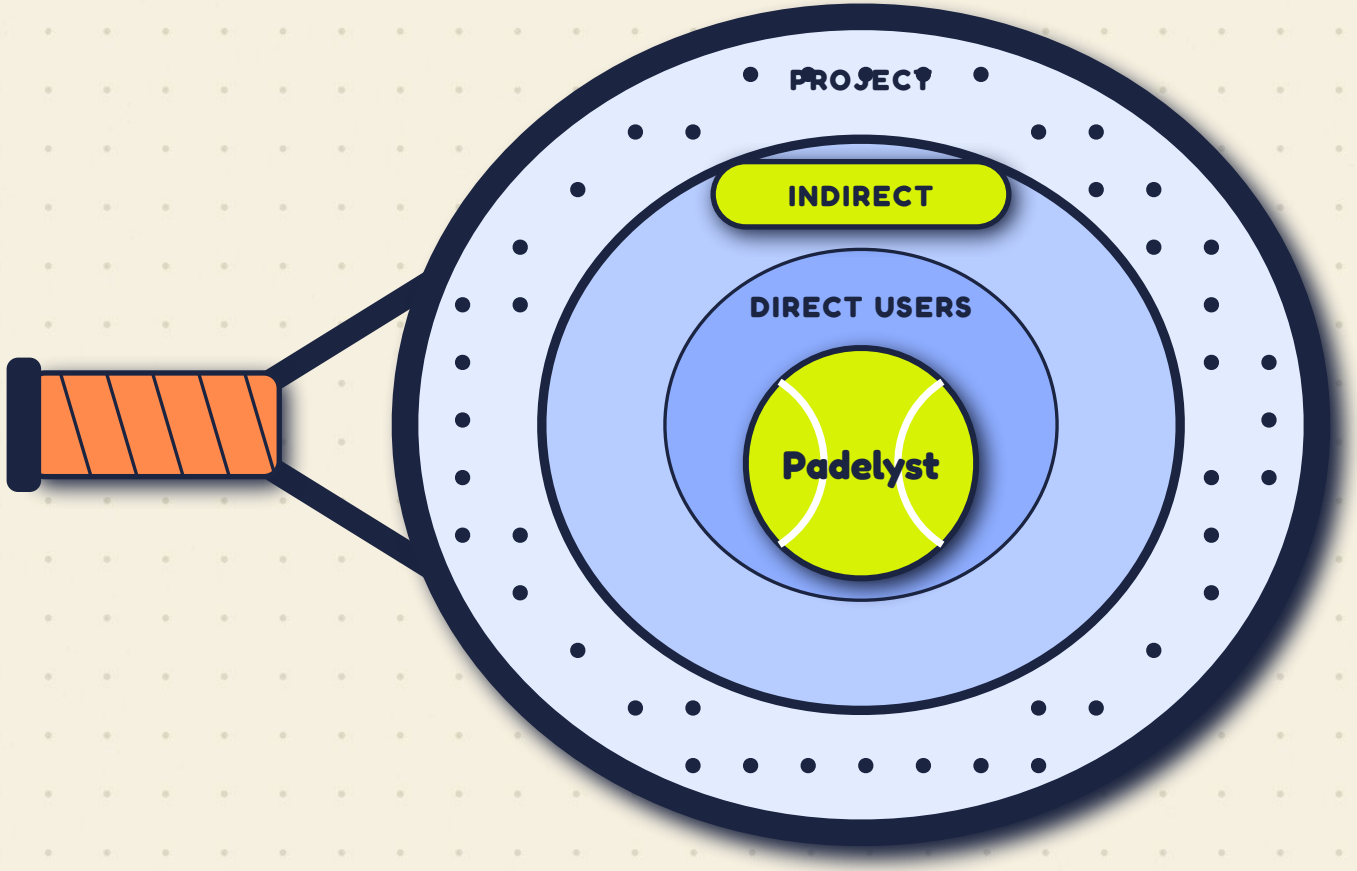
### Coaches
Train from evidence, not memory

### Court owners
A premium service on their courts

### Performance analysts
Stop tagging shots by hand

STAKEHOLDERS

# Around them, those who use the statistics or bring us to clubs




## Indirect

**Academies**
Follow development programmes

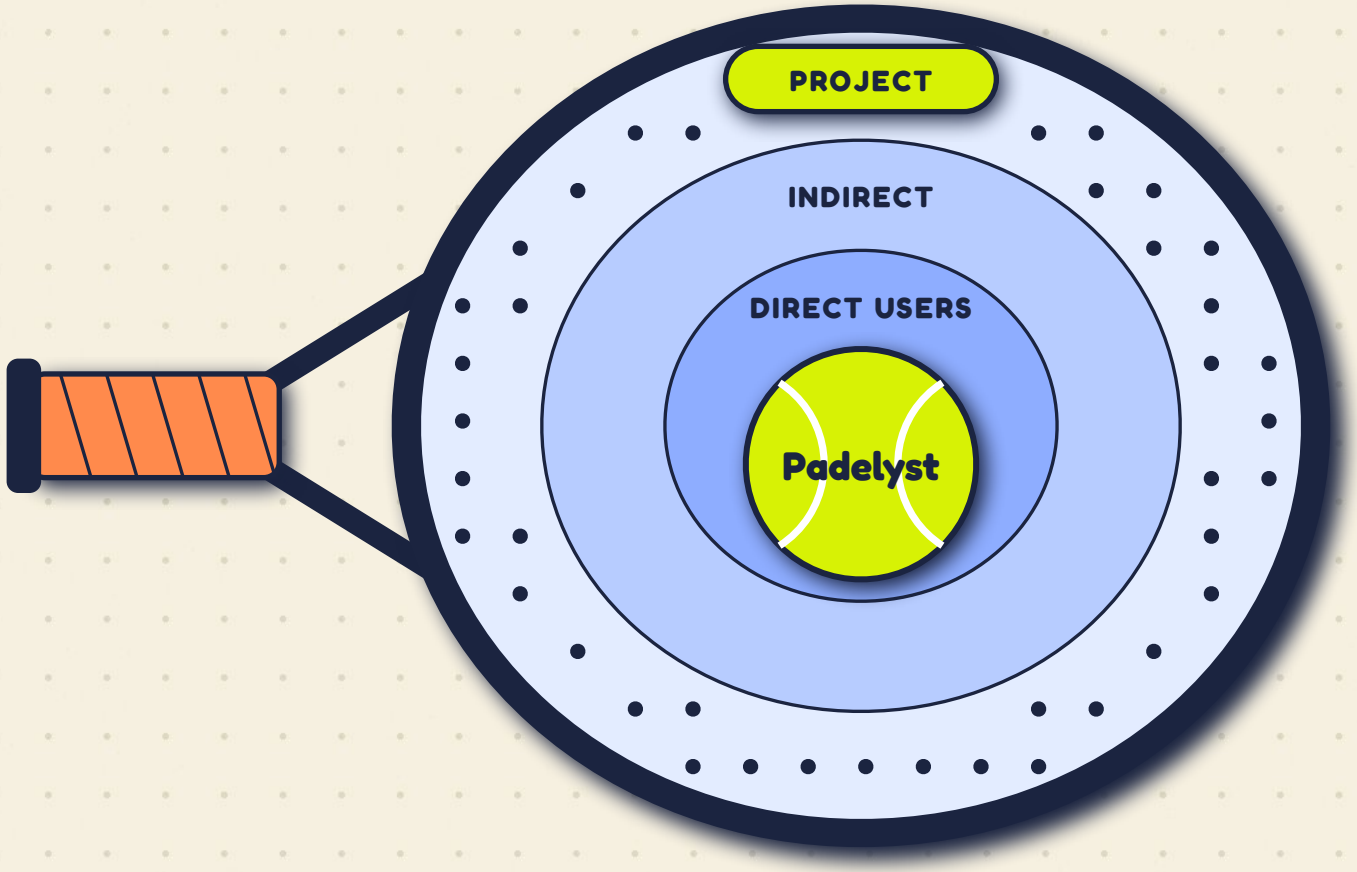
**Federations, tour organisers**
Official statistics

**Broadcasters, media**
Shot statistics for coverage

**Camera vendors, booking platforms**
The channel to reach clubs

STAKEHOLDERS

# And the project itself: mentor and university




**Project**

**LINKS Foundation**
Mentoring and reviews

**Politecnico di Torino**
Course and assessment

PERSONAS

# Giulia wants to improve, but has no time to study her matches



## Giulia, 29

**Amateur player, nurse on shifts**

**PLAYS**
2–3 times a week, lessons with an instructor, club tournaments

**TODAY**
Films on her phone from a bad angle, only for highlights

**PAIN**
Videos fill her phone; no time for statistics

**NEEDS**
Her shot statistics, without rewatching or storing video

Fictional person, built on our use cases. AI-generated photo.

PERSONAS

# Elena coaches fifteen players, with little time for each



## Elena, 38

**Giulia's coach, same club**

**WORKS**
~15 players, one-on-one once a week, a training plan each

**TODAY**
Judges technique only in the little time she has

**PAIN**
Flaws in technique or form go unnoticed

**NEEDS**
A player's shots one after another: type counts, court position

Fictional person, built on our use cases. AI-generated photo.

PERSONAS

# Marco can sell analysis, if it costs him one camera



## Marco, 52

**Owner of the club**

**RUNS**
Several courts, members on monthly or yearly plans

**OFFER**
Camera courts rented at a premium to players who want analysis

**PAIN**
No technical staff to run anything complex

**NEEDS**
One camera with the right framing, plus the engine

Fictional person, built on our use cases. AI-generated photo.

IMPACT

# We contribute to two Sustainable Development Goals




## Good health and well-being

Clear feedback on their own game gives players a reason to keep playing, and keep moving.

**Target 3.4: promote mental health and well-being**




## Industry, innovation and infrastructure

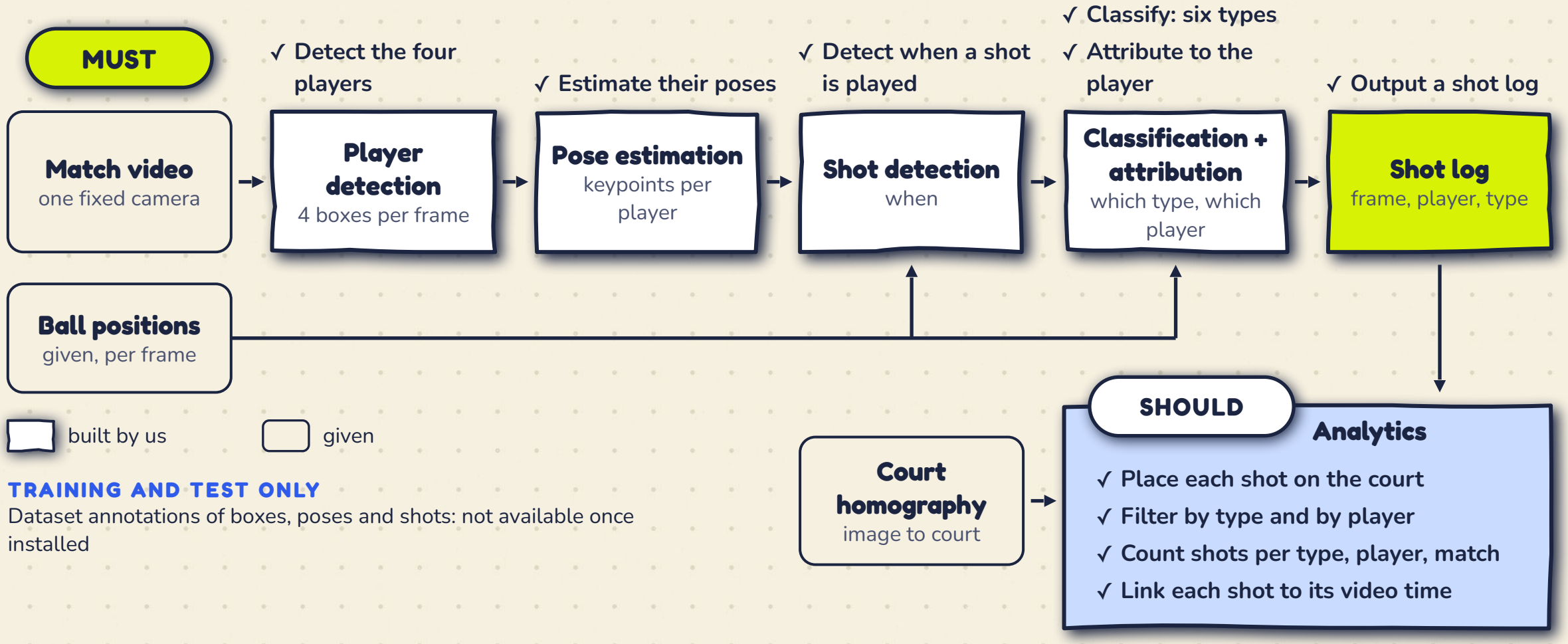
The first reference for shot recognition in padel: open data, publishable code and method.

**Target 9.5: enhance research and technological capabilities**

Source: United Nations, Transforming our world: the 2030 Agenda for Sustainable Development.

REQUIREMENTS

# Every block we build is a requirement




**TRAINING AND TEST ONLY**

Dataset annotations of boxes, poses and shots: not available once installed

REQUIREMENTS

# Offline, reproducible, and robust to our own mistakes

## Reproducible

Same split, same script, every time

## Offline

A whole match processed, no manual step

## Robust

Survives the mistakes of our own detectors

## Light

Stores frames and labels, no images

REQUIREMENTS

# What could come later, and what we won't build

## COULD

- ✓ A confidence score per shot
- ✓ Export the log as a table

## WON'T

- ✓ Real time: live broadcasts already have far more advanced systems
- ✓ User interface

**Our target is the local club, with a small budget: nobody there needs statistics live.**

KPIs

# Two numbers we answer to: when a shot happens, and which one it is

## WHEN

TARGET 0.80

**F1**

on shot events

**67%**

of frames contain no shot

Precision alone rewards silence; recall alone rewards calling "shot" on every frame. F1 balances the two.

## WHICH

TARGET 0.75

**F1**

class-weighted

**5 of 937**

shots are dropshots: rare classes must still count

Class weights set with LINKS from the cost of each error for our users, before results.

**No baseline exists.** No reference results for padel shot recognition: whatever we reach will be the first.

RISKS

# What could go wrong, and how we plan to handle it

| | RISK | MITIGATION |
|---|---|---|
| 1 | **Errors add up along our chain**<br>A wrong box gives a wrong pose, and a missed shot | Check every block on the annotations, to see where errors start |
| 2 | **Players leave the court, glass reflects them**<br>A player hitting outside, or a reflection that looks like one | Track four identities over time; use the court geometry |
| 3 | **Who hit it?**<br>Two players close to the ball | Combine the ball's change of direction with each player's pose |
| 4 | **Rare and noisy labels**<br>Dropshot: 5 of 937 shots; 13 shots shorter than 5 frames | Report every class, never hide one; write down each cleaning rule |
| 5 | **Only professional matches, few shots**<br>937 shots from two finals; no amateur ground truth to test on | A known obstacle from day one: transfer to amateur clubs may suffer, and we state it |

Label counts: our analysis of the PadelTracker100 labels.

adelyst

# Every shot found.
# Every shot named.
# No more rewinding.

**Thank you. Questions?**

**SOURCES**

Dataset: PadelTracker100, Zenodo, CC BY 4.0, doi 10.5281/zenodo.17020011

Courts: FIP, World Padel Report 2026; FIP Research Dept., via Quotidiano Piemontese, 29 Sep 2025

Sustainable Development Goals: United Nations, 2030 Agenda

Project scope: LINKS Foundation, Shot Recognition in Padel Matches Videos

All other figures: our analysis of the dataset labels

**Alessandro de Stasi, Gianbattista Busonera, Gabriele Tebano | Mentor: Giacomo Blanco, LINKS Foundation**